

GLOBE CREATOR

Python Programming Assignment

Topic	Advanced If-Else	Instructor	Pratik Ramteke
Level	Beginner → Intermediate	Subject	Python

Instructions

1. Write a separate Python program for each question.
2. Use **if**, **elif**, **else**, logical operators, and nested conditions where appropriate.
3. Take input from the user using **input()** whenever required.
4. Test your program with different inputs, including edge cases.
5. Use meaningful variable names and proper Python indentation.

Part A — Multiple Conditions

Question 1: Positive, Negative or Zero

Take a number and determine whether it is positive, negative, or zero.

Question 2: Largest of Three Numbers

Take three numbers and find the largest number using **if-elif-else**.

Question 3: Smallest of Three Numbers

Take three numbers and find the smallest number.

Question 4: Number Classification

Take a number and determine whether it is **Positive Even**, **Positive Odd**, **Negative Even**, **Negative Odd**, or **Zero**.

Question 5: Marks and Grade

Take marks and display: **90–100 → A+**, **80–89 → A**, **70–79 → B**, **60–69 → C**, **40–59 → D**, **Below 40 → Fail**.

Part B — Logical Operators

Question 6: Login System

Ask for username and password. If both are correct, display **Login Successful**; otherwise display **Invalid Username or Password**.

Question 7: Discount Calculator

Take shopping amount. Give **20%** discount for **■10,000** or above, **10%** for **■5,000–■9,999**, and no discount below **■5,000**. Display the final amount.

Question 8: Admission Eligibility

A student is eligible if **marks ≥ 60 AND age ≥ 18**. Check and display the result.

Question 9: Employee Bonus

An employee receives a bonus if **experience** $\geq$ 5 years **AND** **salary** $<$ ■50,000.

Question 10: Driving Eligibility

Take age and whether the person has a valid license (**yes/no**). Allow driving only when age is at least 18 and the license is yes.

Part C — Nested If

Question 11: ATM Withdrawal

Ask for account balance and withdrawal amount. The amount must be greater than 0 and must not exceed the balance. If valid, display the remaining balance.

Question 12: Movie Ticket Pricing

Take age and calculate: **Below 5** $\rightarrow$ **Free**, **5–12** $\rightarrow$ ■100, **13–59** $\rightarrow$ ■200, **60+** $\rightarrow$ ■120.

Question 13: Electricity Bill

Calculate: **First 100 units** $\rightarrow$ ■5/unit, **101–200** $\rightarrow$ ■7/unit, **Above 200** $\rightarrow$ ■10/unit. Display the total bill.

Question 14: Simple ATM

Ask the user to select: **1. Check Balance**, **2. Deposit**, **3. Withdraw**. Perform the selected operation using if-elif-else.

Question 15: Online Shopping

Take product price and membership status. If the customer is a member and purchase is above ■5,000, give a 20% discount.

Part D — Challenge Questions

Question 16: Leap Year Checker

Write a program to determine whether a given year is a leap year.

Question 17: Triangle Validator

Take three sides and check whether they can form a triangle.

Question 18: Triangle Type

Take three sides and determine whether the triangle is **Equilateral**, **Isosceles**, or **Scalene**.

Question 19: Password Strength

Ask for a password. Check whether it has at least 8 characters, a number, and uppercase/lowercase characters. Display whether it is **Strong** or **Weak**.

Question 20: Mini Result System

Take marks for Python, SQL, and Statistics. Calculate average: $\geq 80 \rightarrow$ **Excellent**, $\geq 60 \rightarrow$ **Good**, $\geq 40 \rightarrow$ **Pass**, otherwise **Fail**.

Part E — Bonus Challenge

Question 21: ATM Security System

Ask for a PIN. Correct PIN → ask for withdrawal amount. Wrong PIN → **Incorrect PIN**. Correct PIN + sufficient balance → process withdrawal. Correct PIN + insufficient balance → **Insufficient Balance**.

Student Submission Checklist

■	All 21 questions attempted
■	Programs tested with multiple inputs
■	Logical and nested conditions used correctly
■	Code is properly indented
■	Variable names are meaningful

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